Tutorial

problem ①

$$\left.\begin{array}{l} P_1, P_2, \dots, P_n \\ \hline (q \rightarrow r) \end{array}\right\} \text{valid}$$

$$(P_1 \wedge P_2 \dots P_n) \rightarrow (q \rightarrow r)$$

$$\equiv \neg(P_1 \wedge P_2 \wedge \dots \wedge P_n) \vee (q \rightarrow r)$$

$$\equiv \neg(P_1 \wedge P_2 \wedge \dots \wedge P_n) \vee (\neg q \vee r)$$

$$\equiv [\neg(P_1 \wedge P_2 \wedge \dots \wedge P_n) \vee \neg q] \vee r$$

De Morgan $\quad \neg(P_1 \wedge P_2) \equiv \neg P_1 \vee \neg P_2$

$$\equiv \neg(P_1 \wedge P_2 \wedge \dots \wedge P_n \wedge q) \vee r$$

$$\equiv (P_1 \wedge P_2 \wedge \dots \wedge P_n \wedge q) \rightarrow r$$

$\Rightarrow$ argument

$$\begin{array}{l} P_1 \\ P_2 \\ \vdots \\ P_n \\ q \\ \hline \therefore r \end{array}$$

problem (27)

p : I play hockey
q : I am sore
s : I use whirlpool

$p \rightarrow q$
$q \rightarrow s$
$\neg s$

$\} \; p \rightarrow s \equiv \neg s \rightarrow \neg p$

$\neg s \rightarrow \neg p$
(¬s)

$\therefore (\neg p)$

$\left\{ \begin{array}{l} p \rightarrow q \\ p \end{array} \right\}$

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q

$[(p \rightarrow q) \wedge p] \rightarrow q$

$P(x)$: x is an insect
$q(x)$: x has six legs
$S(x,y)$: x eat y.

Domain : Creatures

$\forall x [P(x) \rightarrow q(x)]$

D = organisms
SP = sparrow

$P(D)$
$\neg q(SP)$
$S(SP, D)$

Let a be an arbitrary element

$\left\{ \begin{array}{l} P(a) \rightarrow q(a) \\ P(D) \end{array} \right\} \Rightarrow (q(D))$

$$\begin{cases} P(a) \longrightarrow q(a) \\ P(D) \\ \neg q(sp) \\ S(sp, D) \end{cases} \Rightarrow \underline{q(D)}$$

$$\begin{cases} P(a) \longrightarrow q(a) \equiv \neg q(a) \longrightarrow \neg P(a) \\ \neg q(sp) \end{cases} \Rightarrow \neg P(sp)$$

$$q(D) \wedge \neg P(sp)$$

$$S(sp, D)$$

$$\exists (x, y) \left[S(x, y) \wedge P(y) \right]$$

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problem ⑧

a) $p(x)$: x is a + real

$q(x)$: x^2 is a positive real

$$\forall x \left[P(x) \longrightarrow q(x) \right]$$

$$P(a) \longrightarrow q(a)$$

$$q(a) \longrightarrow P(a)$$

$$\boxed{a = -2}$$

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b) $p(x)$: $x^2 \neq 0$

$q(x)$: $x \neq 0$

$$\forall x \left[P(x) \longrightarrow q(x) \right]$$

$\forall x [P(x) \longrightarrow q(x)]$

$P(a) \longrightarrow q(a)$.

problem (21)

$$\begin{cases} \forall x\,(P(x) \longrightarrow (Q(x) \wedge S(x)) \\ \forall x\,(P(x) \wedge R(x)) \end{cases}$$

$$\forall x\,(R(x) \wedge S(x))$$

Let (a) be an (arbitrary) element.

- $P(a) \longrightarrow (Q(a) \wedge S(a))$
- $P(a) \wedge R(a)$

$P(a) \wedge R(a)$

$P(a)$
$R(a)$

$P(a)$
$P(a) \longrightarrow (Q(a) \wedge S(a))$

$Q(a) \wedge S(a)$

$P(a)$
$R(a)$
$Q(a) \wedge S(a)$

$P(a)$
$R(a)$
$Q(a)$
$S(a)$

$(P(a) \wedge S(a))$

$(R(a) \wedge S(a))$

$\Rightarrow \quad \forall x\,[R(x) \wedge S(x)]$

problem ⑤

$\forall x\,[P(x) \vee Q(x)]$

$\forall x\,[\neg Q(x) \vee S(x)]$

$\forall x\,[R(x) \longrightarrow \neg S(x)]$

$\exists x\, \neg P(x)$

$\exists x\, \neg R(x)$

choose element "c" such that $\neg P(c)$ is true.

$P(c) \vee Q(c) \equiv \neg P(c) \longrightarrow Q(c)$

$\neg Q(c) \vee S(c) \equiv Q(c) \longrightarrow S(c)$

$R(c) \longrightarrow \neg S(c) \equiv S(c) \longrightarrow \neg R(c)$

$\neg P(c)$

$\neg P(c) \longrightarrow \neg R(c)$

$\neg p(c) \longrightarrow \neg R(c)$

$\neg p(c)$

$\neg R(c)$

$\Rightarrow \boxed{\exists x \quad \neg R(x)}$

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problem (6)

def: $n \in \mathbb{Z}^+$ n is odd if $n = 2m+1$
$m \in \mathbb{Z}^+$

$m = p^2 - l^2 \qquad p, l \in \mathbb{Z}^+$

if p, l both are even, (X)

$p = 2k \quad , \quad l = 2q$

$p^2 - l^2 = (2k)^2 - (2q)^2 = 4k^2 - 4q^2$

$= 2[2k^2 - 2q^2]$

if p, l both are odd (X)

$p = 2k+1 \quad , \quad l = 2q+1$

$p^2 - l^2 = (2k+1)^2 - (2q+1)^2$

$= 4k^2 + 4k + 1 - 4q^2 - 4q - 1$

$= 2[2k^2 + 2k - 2q^2 - 2q]$

if p is odd L is even

$p = 2k+1$ $L = 2q$

$p^2 - L^2 = 4k^2 + 4k + 1 - 4q^2$

$= 2[2k^2 + 2k - 2q^2] + 1$

$\Rightarrow n = 2m + 1$

$m = 2k^2 + 2k - 2q^2$

$k = q$

$p = k + 1$, $L = k$

Proof:

Since n is odd $\Rightarrow n = 2m + 1$

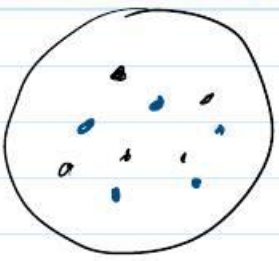
$n = 2m + 1 + 0$

$= (2m + 1 + m^2) - m^2$

$= (m+1)^2 - m^2$

$= p^2 - L^2$

Problem (13)



proof by contradiction

$p \rightarrow q$

- you pick (3) socks
- you did not get ~~any~~ pair.

⇒ 1 2 3

1: black or blue

2: black or (blue)

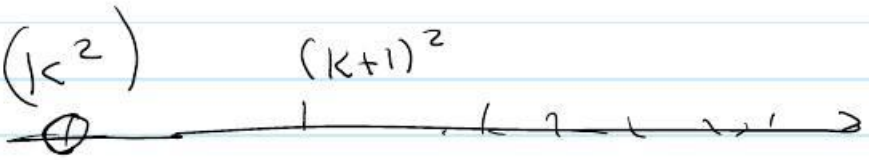
3: must be ~~neither~~

problem (7)

n is a perfect square

n+2 is not a perfect square

$\{ n = k^2 \}$



n s

$1^2, 2^2 \to 3^2, 4^2 \ldots$

$(k)^2$ $(k+1)^2$

$(n+2 > n)$

$(n = k^2)$

$(k+1)^2 = k^2 + 2k + 1$

$= n + 2k + 1$

$k = 0$

$n = 1^2 = 1$

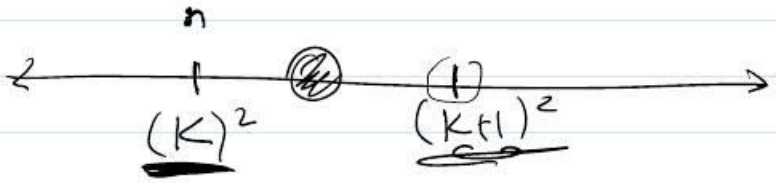
$n+2 = (3)$ ⊗

$k > 0$

$n + 2k + 1 > n + 2$

$n+2 = (3)^1$ ✗

$(k+1)^2 > n+2 \qquad k \neq 0$



$n < n+2 < (K+1)^2$

$n+2$ cannot be a perfect square !!

problem (8)

$x \in Q'$, $y \in Q'$

$\Rightarrow \boxed{xy \in Q'}$

$\sqrt{3}$, $\sqrt{5}$, e

$x, y \qquad x = y = \sqrt{3}$

$(\sqrt{3})^2 = 3 \in Q$

problem (9)

Let $\left(\frac{a}{b}\right) \in Q$ s.t $b \neq 0$ let $x \in Q'$

$\left(\frac{a}{b} x\right) \in Q'$?!

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if $\frac{a}{n}, \frac{c}{i} \in Q$ s.t $b, d \neq 0$

if $\frac{a}{b}, \frac{c}{d} \in Q$ s.t $b, d \neq 0$

$\frac{a}{b} \times \frac{c}{d} \in Q$

$\frac{a}{b} \in Q \quad , x \in Q'$

$\Rightarrow \frac{b}{a} \in Q$

$\left(x \frac{a}{b}\right)\left(\frac{b}{a}\right) = (x)$

irrational

problem (10)

$x \in Q' \Rightarrow \frac{1}{x} \in (Q')$

$\frac{1}{x} \in Q \longrightarrow x \in Q$

let $\frac{1}{x} \in \frac{a}{b}$ $b \neq 0$ $\Rightarrow x = \left(\frac{b}{a}\right) \in Q$

problem (16)

$n \in Z^+$

n odd $\iff$ $5n+6$ odd

$(n \text{ odd} \longrightarrow 5n+6 \text{ odd}) \wedge (5n+6 \text{ odd} \longrightarrow n \text{ is odd})$

$(n \text{ odd} \rightarrow 5n+6 \text{ odd}) \wedge (5n+6 \text{ odd} \rightarrow n \text{ is odd})$

$\Rightarrow$ $n = 2m+1$

$5n+6 = 5(2m+1)+6 = 10m + 5 + 6$

$= 10m + 5 + 5 + 1 = 10m + 10 + 1$

$= 2(5m+5) + 1 \Rightarrow$

$5n+6$ is odd

$\Leftarrow$ $5n+6$ is odd $\rightarrow$ n is odd

n is even $\rightarrow$ $5n+6$ is even

$n = 2m$

$5n+6 = 5(2m) + 6 = 10m + 6$

$= 2(5m+3)$

$5n+6$ is even